Closing the Transitivity Gap: A Degree-Preserving Configuration Model with Triangle Stubs for the Amazon Co-Purchase Network

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1. Problem and Motivation

On the Week 4 scorecard, no random graph model reproduces both the degree sequence and the triangle structure of the Amazon co-purchase network (SNAP com-Amazon: 334,863 nodes, 925,872 edges, 667,129 triangles). The configuration model matches the degrees exactly but has a transitivity below 0.001; Holme–Kim creates triangles, but its maximum degree is 5,647 instead of 549 and its transitivity is 0.032 instead of 0.205. Triangles matter because they change dynamics such as epidemic spreading (Miller, 2009): a synthetic graph without them is not similar where it matters.

2. Research Question

To what extent can a triangle-based configuration model reproduce the clustering structure of the Amazon co-purchase network while preserving its degree sequence? Targets: transitivity 0.205 and average clustering 0.397, with every degree kept exactly.

3. Method

We implement the configuration model with triangle stubs (Newman, 2009). For each node *i* we read its degree k_i and its number of triangles Δ_i from Amazon and assign $t_i = \min(\Delta_i, \lfloor k_i/2 \rfloor)$ triangle stubs and $s_i = k_i - 2t_i$ edge stubs. Edge stubs are shuffled and paired; triangle stubs are shuffled and grouped in threes, each group forming a triangle. To keep every degree exact, invalid groups (repeated node or existing edge) are redrawn instead of deleted; groups that cannot be placed become two edge stubs, and remaining conflicts are resolved by degree-preserving swaps. The generator uses packed NumPy arrays. A K4 extension (groups of four) will target overlapping triangles.

4. Prediction Before Building

Because triangles are placed on purpose and degrees do not change, the reachable transitivity can be computed in advance: $T_{max} \approx \Sigma t_i \,/\, \Sigma \tfrac{1}{2} k_i(k_i - 1)$. For Amazon, $T_{max} = 0.077$, i.e. 37.5 % of the real value, because 56 % of nodes lie in more triangles than $\lfloor k_i/2 \rfloor$ allows. The gap is concentrated at low degrees (degree 5: real clustering 0.509, model at most 0.185), pointing to dense groups such as K4. The bound applies to this basic model only; accidental triangles are negligible (configuration model: 0.00005).

5. Evaluation and Baselines

Success criteria are fixed before building: transitivity and average clustering within ±5 % of the prediction (0.077 and 0.205), 100 % of nodes with their exact degree, clustering by degree matching the real curve below the cap, and an SIR epidemic closer to Amazon than the configuration model (at least 50 runs). Average path length is reported but not targeted.

Baselines are the configuration model, Chung–Lu and Holme–Kim, measured with the same script on at least five graphs per model (mean ± standard deviation).

6. Software Stack

Python 3; NumPy for the generator; networkit for large-scale measurements; NetworkX for baselines; SciPy and Matplotlib for statistics and plots. Large runs use the university cluster (SCC); code and report are kept on git.sbg.ac.at.

7. Cost Analysis for 10^9 Nodes

Assuming Amazon's density (about 2.77×10^9 edges, 4-byte ids) and the Week 5 method: peak memory is about 100 GB (CSR 30 GB, stubs 22 GB, duplicate-check hash set 44 GB, per-node counts 4 GB); runtime is about 15 minutes on one core, dominated by about 8.3×10^9 random accesses at 100 ns (64 cores: at most 15× faster by Amdahl's law); the hardware costs about €11,700 (128 GB RAM €4,480, 64 cores €7,040, 1 TB SSD €160; prices from Week 5). One SCC node with 2.3 TB of RAM suffices, so we scale up rather than out. Preliminary prototype measurements (10^7 nodes in 30 s; about 80 bytes per edge) will be used to validate and refine these estimates.

8. Plan and Risks

November: parameter extraction, generator and baselines. December: K4 extension, scaling measurements and evaluation. January: report and presentation (due 26 January 2027). Main risks are a remaining large gap (addressed by K4 stubs), memory at 10^9 nodes (measure at 10^5–10^7 first, then optimise) and noisy SIR results (at least 50 runs).

References

Newman, M. E. J. (2009). Random graphs with clustering. Phys. Rev. Lett. 103, 058701. · Miller, J. C. (2009). Percolation and epidemics in random clustered networks. Phys. Rev. E 80, 020901. · Holme, P., Kim, B. J. (2002). Growing scale-free networks with tunable clustering. Phys. Rev. E 65, 026107. · Yang, J., Leskovec, J. (2012). SNAP com-Amazon dataset.